

NFC API

Development Documentation

V1.1.0

Update Records

Date	Description
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1 Introduction

Package: com.common.apiutil.nfc

1.1 SelectCardReturn Class

Class	Description
Nfc	Reflection interface
NfcUtil	MCU
SelectCardReturn	Card information

1.1.1 Card type enumeration

```
/**
 * @Function Card type enumeration
 *
 * @Parameter
 * 1.cardType: M0,M1,ISO15693,FELICA,ID_CARD,TYPE_A,TYPE_B
 *
 */

public class SelectCardReturn {
    private CardTypeEnum cardType;
    private byte[] cardNum;
    public SelectCardReturn() {

    }

    public void setCardType(CardTypeEnum cardType) {
        this.cardType = cardType;
    }

    public CardTypeEnum getCardType() {
        return this.cardType;
    }
}
```

1.2 NfcUtil Class

Supports identification of CPU card and M1 card

1.2.1 Get Instance

```
/**
 * @Function Get instance
```

```

*
* @Parameter context
*
* */

public synchronized static final NfcUtil getInstance(Context context)

```

1.2.2 Initialize the serial port

```

/**
 * @Function Initialize the serial port
 *
 * */

public synchronized void initSerial()

```

1.2.3 Destroy the serial port

```

/**
 * @Function Destroy the serial port
 *
 * */

public synchronized void destroySerial()

```

1.2.4 Check the microcontroller software version

```

/**
 * @Function Check the microcontroller software version
 *
 * @Return Microcontroller software version number
 *
 * */

public synchronized int checkVersion()

```

1.2.5 CPU Card-Select Card

```

/**
 * @Function CPU Card-Select Card
 *
 * */

public synchronized SelectCardReturn selectCard()

```

1.2.6 CPU Card-Send the APDU Command

```
/**
 * @Function CPU Card-Send the APDU Command
 *
 * */

public synchronized byte[] sendAPDU(byte[] cmd, int timeout)
```

1.2.7 M1-Authentication

```
/**
 * @Function M1-Authentication
 *
 * @Exception WrongParamException
 *
 * */

public synchronized boolean M1_AUTHENTICATE(int block, int keyType, byte[] key)
throws WrongParamException
```

1.2.8 M1-Read Block

```
/**
 * @Function M1-Read Block
 *
 * */

public synchronized byte[] M1_READ_BLOCK(int block) throws WrongParamException
```

1.2.9 M1-Write Block

```
/**
 * @Function M1-Write Block
 *
 * @Exception WrongParamException
 *
 * */

public synchronized boolean M1_WRITE_BLOCK(int block, byte[] data) throws
WrongParamException
```

1.2.10 M1-Close the Card

```
/**
 * @Function M1-Close the Card
 *
 * */

public synchronized boolean M1_CLOSE_CARD()
```

1.2.11 M1-Initialize the e-wallet

```
/**
 * @Function M1-Initialize the e-wallet
 *
 * @Parameter
 * 1.block: Write Block:0 - 255
 * 2.money: Value
 *
 * @Return True, success; false, failed
 *
 * @Exception WrongParamException
 *
 * */

public boolean M1_INIT_WALLET(int block, int money) throws WrongParamException
```

1.2.12 M1-Read the e-wallet

```
/**
 * @Function M1-Read the e-wallet
 *
 * @Parameter
 * 1.block: Write Block:0 - 255
 *
 * @Return Value
 *
 * @Exception WrongParamException, ReadWalletMoneyWrongException,
NotInitWalletException
 *
 * */

public Long M1_READ_WALLET(int block) throws WrongParamException,
ReadWalletMoneyWrongException, NotInitWalletException
```

1.2.13 M1-Write the e-wallet

```

/**
 * @Function M1-Write the e-wallet
 *
 * @Parameter
 * 1.block: Write Block:0 - 255
 * 2.money: Value
 * 3.dealType: Type:0, Add;1, Sub
 *
 * @Return True, success; false, failed
 *
 * @Exception WrongParamException
 *
 * */

public boolean M1_WRITE_WALLET(int block, int dealType, int money) throws
WrongParamException

```

Type	Description
WrongParamException	Parameter error exception
ReadWalletMoneyWrongException	Read e-wallet error exception
NotInitWalletException	E-wallet is not initialized exception
NoClassObjectsException	The required class was not found

1.3 NFC Model

1.3.1 Supported Card Types

It depends on the specific NFC chip and terminal device

1.3.2 Mifare UltraLight EV1 card

a Write Block

```

/**
 * @Function Write Block
 *
 * @Parameter
 * 1.noBlock: Block
 * 2.inBuf: Data
 * 3.inLen: Length
 *
 * @Exception CommonException
 *
 * */

```



```
public synchronized void ultraigh_write_page(byte noBlock, byte[] inBuf, int
inLen) throws CommonException
```

b Read Block

```
/**
 * @Function Read Block
 *
 * @Parameter
 * 1.noBlock: Block
 *
 * @Exception CommonException
 *
 * */

public synchronized byte[] ultraigh_read_page(byte noBlock) throws CommonException
```

c Ultraigh EV1 Password Verification

```
/**
 * @Function Ultraigh EV1 Password verification
 *
 * @Parameter
 * 1.noBlock: 0-255
 * 2.pwd: A 4-byte password
 *
 * @Return success : 2 bytes. Failed : exception
 *
 * @Exception CommonException
 *
 * */

public synchronized byte[] ultraigh_pwd_auth(byte noBlock, byte[] pwd) throws
CommonException
```

1.3.3 Mifare Desfire EV2/EV3 Card

[./NFC/Defire_Ev2&Ev3_Card.pdf](#)

1.3.4 CPU Card

[./NFC/cpu_card_api.pdf](#)

1.3.5 Mifare Classic Card

[./NFC/mifare_classic_card_api.pdf](#)

1.3.6 Felica Card

[./NFC/felica_card_api.pdf](#)

1.3.7 Mifare Desfire EV1 Card

[./NFC/desfire_ev1_card_api.pdf](#)